Question 1 **4 marks**

Define and explain the following terms

(i) Isotope

(ii) Half life

(iii) Beta irradiation

(iv) Alpha decay

[1 + 1 + 1 + 1 = 4 marks]

(i) Isotope ; atoms with same number of protons but different number of neutrons

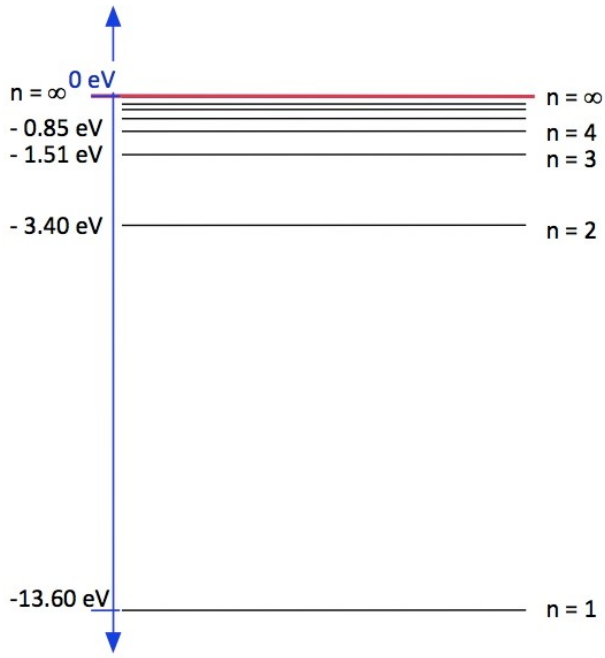
(ii) Half life: time taken for half the radioactive nuclei to decay.

(iii) Beta irradiation: high energy electrons emitted during radioactive decay

(iv) Alpha decay: decay of a nucleus with the emission of an alpha particle with is a he nucleus of 2 protons and 2 neutrons.

Question 2 **7 marks**

The diagram shows the energy of the ground state of the hydrogen atom, as well as the energies of three of its excited states.



(a) A hydrogen atom absorbs radiation so that it is excited to the third energy level above the ground state. What are all of the possible energies, in eV, of photons that could then be emitted by the hydrogen atom?

Excited to n = 4 level, so the transitions are:

n = 4 to n = 3: E_{photon} = 1.51 – 0.85 = 0.66 eV then n = 3 to n = 2: E_{photon} = 3.40 – 1.51 = 1.89 eV

n = 4 to n = 2: E_{photon} = 3.40 – 0.85 = 2.55 eV then n = 2 to n = 1: E_{photon} = 13.60 – 3.40 = 10.20 eV

n = 4 to n = 1: E_{photon} = 13.60 – 0.85 = 12.75 eV

(b) Suppose in a sample of hydrogen atoms at room temperature the atoms are all in the ground state. Justify the statement that none of the radiation absorbed by any of these hydrogen atoms is in the visible part of the electromagnetic spectrum.

The smallest quantity of energy that can be absorbed by an atom in the ground state is:

13.60 eV – 3.40 eV = 10.20 eV

The wavelength of this radiation is given by:

$\lambda = hc/E = 6.626 \times 10^{-34}$ Js x 2.998×10^{8} m/s / [10.20 eV x 1.602×10^{-19} J/eV]

$= 1.216 \times 10^{-7}$ nm which is in the ultraviolet.

Since all other radiation will result from greater energy changes for the atom, the radiation will all have shorter wavelengths and will be even further into the UV, i.e. no visible light

[3 + 4 = 7 marks]

Question 3 **19 marks**

Americium-241, $^{241}_{95}Am$, was one of the radioactive sources used to demonstrate the properties of radiation during a PHYC10006 lecture. The source was contained in a smoke detector, and was a 3.8 x 10^4 Bq source of radiation.

(a) In the lecture it was shown that when a Geiger tube was placed close to the smoke detector it detected radiation – a clicking sound was observed. Then, when the Geiger tube was pulled back so that it was a centimetre away from the source, the rate of detection of radiation decreased dramatically.

What sort of nuclear radiation was emitted by the $^{241}_{95}Am$: α, β, or γ? Use the observed behaviour to justify your answer.

α-radiation is emitted

The other forms of radiation would both penetrate through the centimeter of air, so since this radiation cannot penetrate through even a centimeter of air (as seen by the drop in count rate) this cannot be β, or γ and must be α.

(b) Radon is a gas that decays via the same process as $^{241}_{95}Am$. Measurements have shown that the average concentration of radon in Australian homes is approximately 11 Bq/m^3, much lower than in most other countries. However, as it is possible for there to be variation from home to home, the Australian Radiation Protection and Nuclear Safety Agency recommends that people seek advice about how they can reduce the radon in their homes if its activity density exceeds 200 Bq/m^3.

Why is this concentration of radon a concern, while the much more active 3.8 x 10^4 Bq $^{241}_{95}Am$ sources in smoke detectors are not considered to be a risk to health?

α-radiation cannot travel far through air, or through paper, because it is highly ionising.
Consequently when an α-emitter is outside the body, e.g. the source in the smoke detector, it poses minimal risk as the probability of it ionising within the body is low – it does not enter body tissue.
However, because radon is a gas is can be breathed in.
When inside the body its highly ionising character means it ionises inside the body and results in risk.

The table below provides the relative atomic masses (in u) of several isotopes with similar mass numbers to $^{241}_{95}Am$.

$^{241}_{97}Bk$ 241.06023	$^{242}_{97}Bk$ 242.06198	$^{243}_{97}Bk$ 243.06301	$^{244}_{97}Bk$ 244.06518	$^{245}_{97}Bk$ 245.06636
$^{240}_{96}Cm$ 240.05553	$^{241}_{96}Cm$ 241.05765	$^{242}_{96}Cm$ 242.05884	$^{243}_{96}Cm$ 243.06139	$^{244}_{96}Cm$ 244.06275
$^{239}_{95}Am$ 239.05302	$^{240}_{95}Am$ 240.05530	$^{241}_{95}Am$ **241.05682**	$^{242}_{95}Am$ 242.05955	$^{243}_{95}Am$ 243.06138
$^{238}_{94}Pu$ 238.04956	$^{239}_{94}Pu$ 239.05216	$^{240}_{94}Pu$ 240.05381	$^{241}_{94}Pu$ 241.05685	$^{242}_{94}Pu$ 242.05874
$^{237}_{93}Np$ 237.04817	$^{238}_{93}Np$ 238.05095	$^{239}_{93}Np$ 239.05294	$^{240}_{93}Np$ 240.05616	$^{241}_{93}Np$ 241.05826

Rest masses of: $^{4}_{2}He$ atom: 4.00260 u; electron: 0.00055 u; positron: 0.00055 u.

(c) (i) Write a balanced equation for the decay process taking place.

$^{241}_{95}Am \rightarrow {}^{237}_{93}Np + {}^{4}_{2}He$

(ii) Use this equation and the data in the table above to provide further evidence that the decay process you chose in (a) is spontaneous for $^{241}_{95}Am$.

The process will be spontaneous if mass decreases
i.e. total mass of products, $^{237}_{93}Np + {}^{4}_{2}He$, is less than total mass of reactants, $^{241}_{95}Am$.
Mass of $^{241}_{95}Am$ = 241.05682 u; Mass of $^{237}_{93}Np + {}^{4}_{2}He$ = 237.04817 u + 4.00260 u = 241.05077 u

Mass has decreased from 241.05682 u to 241.05077 u, therefore the α-decay is spontaneous.

(d) The half-life of $^{241}_{95}Am$ is 432 years. By what factor would the count rate from a smoke detector decrease after 10 years?

R/R_0 = exp (-ln2 *t* /432)
= exp (-0.693 x 10/432)
= 0.984
So, 98 % of the nuclei remain undecayed, and the count rate will also be 98% of the original value.

(e) There is 0.3 x 10^{-6} g of $^{241}_{95}Am$ in the smoke detector. Show that such a source would have an activity of 3.8 x 10^{4} Bq.

Activity, *R* = r N, where r is the decay constant, and N is the number of nuclei

N = mass/atomic mass x Avogadro's number = 0.3 x 10^{-6} g/(241 g/mole) x 6.023 x 10^{23} mol^{-1}

r = ln2/$t_{1/2}$ = 0.693/432 years
R = [0.693/432 years] x [0.3 x 10^{-6} g/(241 g/mole) x 6.023 x 10^{23} mol^{-1}] = 1.2 x 10^{12} decay/yr

Activity is R in Bq which is decays per second = decays per year / seconds in a year
So R = [0.693/432 years] x [0.3 x 10^{-6} g/(241 g/mole) x 6.023 x 10^{23} mol^{-1}]
/ [365 days/yr x 24 hr/day x 3600 s/hr]

= 3.8 x 10^{4} decays per second

[3 + 4 + 5 + 2 + 5 = 19 marks]

Question 4 **11 marks**

The table below provides the relative atomic masses (in u) of several isotopes with similar mass numbers to $^{222}_{86}Rn$. The rest masses of some common particles are also given below the table.

$^{222}_{88}Ra$ 222.01538	$^{223}_{88}Ra$ 223.01850	$^{224}_{88}Ra$ 224.02021	$^{225}_{88}Ra$ 225.02361	$^{226}_{88}Ra$ 226.02541
$^{221}_{87}Fr$ 221.01426	$^{222}_{87}Fr$ 222.01755	$^{223}_{87}Fr$ 223.01974	$^{224}_{87}Fr$ 224.02326	$^{225}_{87}Fr$ 225.02557
$^{220}_{86}Rn$ 220.01139	$^{221}_{86}Rn$ 221.01554	$^{222}_{86}Rn$ **222.01758**	$^{223}_{86}Rn$ 223.02179	$^{224}_{86}Rn$ 224.02409
$^{219}_{85}At$ 219.01243	$^{220}_{85}At$ 220.01476	$^{221}_{85}At$ 221.01559	$^{222}_{85}At$ 222.01784	$^{223}_{85}At$ 223.01914
$^{218}_{84}Po$ 218.00897	$^{219}_{84}Po$ 219.01374	$^{220}_{84}Po$ 220.01660	$^{221}_{84}Po$ 221.0	$^{222}_{84}Po$ 222.0

Rest masses of: $^{4}_{2}He$ atom: 4.002602 u; electron: 0.000549 u; positron: 0.000549 u.

$^{1}_{1}H$ atom: 1.007825 u; neutron: 1.008665 u; proton: 1.007276 u

(a) (i) Write an equation to describe the α-decay of $^{222}_{86}Rn$.

(ii) Use the data above to justify the statement that α-decay of $^{222}_{86}Rn$ is spontaneous.

(b) Calculate the average energy required to remove one nucleon from a nucleus of $^{4}_{2}He$ in an imaginary process in which the nucleus is separated into hydrogen atoms and neutrons. Express your answer in MeV.

(c) The value of the ratio $\frac{number\ of\ neutrons}{number\ of\ protons}$

for both $^{52}_{26}Fe$ (iron-52) and $^{4}_{2}He$ (helium-4, i.e. an α-particle) is 1.0 .

Helium-4 is stable, but iron-52 is radioactive. When iron-52 decays its nucleus captures an inner shell electron. Explain why the same ratio between the numbers of neutrons and protons in the nucleus results in stability for helium-4, but not for iron-52.

$$^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}He$$

If this is spontaneous, then the mass on the right will be less than the mass on the left **[1 mark]**

Mass of initial atom = 222.01758 u

Total mass of products of α-decay is 218.00897 u + 4.002602 u = 222.01157 u

1 mark for setting this up

The final mass is less than the initial mass, and therefore so is the energy. A decrease in energy in the decay indicates that the decay is spontaneous.

A decrease in mass, so this is spontaneous. **[1 mark for correct comparison of masses]**

Mass of helium – 2 mass of hydrogen – 2 mass of neutron = mass defect

[1 mark for statement re what is happening is calculating mass defect]

= 4.002602 - 2 x 1.007825 – 2 x 1.008665 = 0.030378 u **[1 mark]**

Multiply by 931.5 MeV/u: 0.030378 u x 931.5 MeV/u = 28.297 MeV **[1 mark]**

Divide by 4, as 4 nucleons produced: 7.0743 MeV per nucleon **[1 mark]**

Two forces are important in the nucleus:

The electrostatic repulsion between the protons, which obeys an inverse square law, decreases in proportion to $1/r^2$, and operates between all protons in even large nuclei, even though it's weaker between the two sides of a lead-208 nucleus than the two sides of a helium-4 nucleus.
[2 x 1/2 marks –which particles, range]

The strong nuclear force, which acts between all nucleons, whether they are protons or neutrons; this force is short-range, with a range of about 1×10^{-15} m, so it only acts between neighbouring nucleons.
[2 x 1/2 marks – which particles, range]

Now, applying this to the iron versus helium example: As the nucleus becomes larger (iron compared to helium) **the protons that are further away from each other still exert repulsive electrostatic forces [1/2 mark]**, but the attractive component of the potential energy from the attractions between nucleons does not grow so quickly as **only close-neighbours can contribute – all nucleons in He satisfy this, but not in the case of lead [1/2 mark]**, so more neutrons are required as **they add the effects of attractions to close neighbours without adding to the impact of the repulsive forces**. Iron-52 doesn't have enough neutrons for this to occur, so it's radioactive**[1 mark]**

[(1 + 2) + 4 + 4 = 11 marks]

Question 5 — 4 marks

The isotope ${}^{18}_{9}F$ is used in the compound Fluoro-2-Deoxy-D-Glucose (FDG) to produce images of the body using Positron Emission Tomography (PET).

Positrons emitted from the fluorine-18 nuclei in FDG annihilate electrons and produce a pair of X-ray photons that are detected.

Use the data below to *identify* and *explain* one advantage of using a compound involving the ${}^{18}_{9}F$ nuclide in PET imaging, rather than the ${}^{11}_{6}C$ nuclide.

Isotope	Maximum positron range	Half life
${}^{18}_{9}F$	2.6 millimetre	110 minute
${}^{11}_{6}C$	3.8 millimetre	20 minute

Two options here – marks for each underlined idea, to maximum of 4 marks:

1. The positron from the fluorine has a **shorter range before is annihilates an electron** – this means the **photon pairs are being produced close to where the FDG has a high concentration**. When there is a longer range, then a sphere with a radius equal to the maximum range can produce photon pairs, so the **certainty of where the emitter is located is less**. As a result the image has **less resolution** – each point in the object maps onto a blurry sphere on the image, and the **images of adjacent points can overlap – image is not clear**.
2. The half-life of the **fluorine isotope is longer**. This means that the **rate of positrons being emitted by the isotope is higher over a longer period of time**. It is easier to collect images over a longer period – once injected with the isotope there is a **longer time that useful positron intensity exists**, and also the **time taken to transport the radioactive isotope** to where the imaging is being done is less critical

4 marks

Question 6 **9 marks**

(a) (i) Explain why the existence of a cutoff frequency in the photoelectric effect more strongly favours a particle theory rather than a wave theory of light.

(ii) If the threshold wavelength in the photoelectric effect increases when the emitting metal is changed to a different metal, what can you say about the work function of the two metals?

(b) Two light sources are used in a photoelectric effect experiment to determine the work function of a particular metal surface. When green light from a mercury lamp ($\lambda = 546.1\,\text{nm}$) is used, a retarding potential of 1.62V reduces the photocurrent to zero.

(i) Based on this measurement, what is the work function for this metal?

(ii) What stopping potential would be observed when using the yellow light from a helium discharge tube ($\lambda = 587.5\,\text{nm}$)?

[(3 + 2) + (2 + 2) = 9 marks]

(a) (i) According to the wave theory, light of any frequency can cause electrons to be ejected as long as the light is intense enough. A higher intensity corresponds to a greater electric field magnitude and more energy. Therefore, there should be no frequency below which the photoelectric effect does not occur. According to the particle theory, however, each photon carries an amount of energy which depends upon its frequency. Increasing the intensity of the light increases the number of photons but does not increase the energy of the individual photons. The cutoff frequency is that frequency at which the energy of the photon equals the work function. If the frequency of the incoming light is below the cutoff, then the electrons will not be ejected because no individual photon has enough energy to eject an electron.

(ii) If the threshold wavelength increases for the second metal, then the second metal has a smaller work function than the first metal. Longer wavelength corresponds to lower energy. It will take less energy for the electron to escape the surface of the second metal.

(b) (i)

$$\text{eV}_{\text{stop}} = hf - \phi \quad \text{so} \quad \phi = hf - \text{eV}_{\text{stop}} = \frac{hc}{\lambda} - \text{eV}_{\text{stop}} = \frac{4.136\times10^{-15}\times3\times10^{8}}{546.1\times10^{-9}} - 1.62 = 2.27 - 1.62 = 0.65\,eV$$

(ii) $$\text{eV}_{\text{stop}} = \frac{hc}{\lambda} - W_0 = \frac{4.136\times10^{-15}\times3\times10^{8}}{587.5\times10^{-9}} - 0.65 = 2.11 - 0.65 = 1.46\,V$$